Freight Rate Prediction

Machine Learning Assessment Report

Objective	Predict posted freight rates from shipment, lane, market, quote, and date information.
Development data	48,000 labeled loads
Final validation data	12,000 unlabeled loads
Development period	January–October 2025
Final prediction period	November–December 2025
Primary validation split	January–September → October 2025
Selected model	Log-target HistGradientBoostingRegressor
Validation MAE	$115.45
Validation RMSE	$647.40
Validation R^2	0.8206

Note: The validation metrics above are development-set results from the chronological October 2025 holdout. The final November–December evaluation is performed by Spotter after submission.

1. Assessment Objective

This is a supervised regression problem. The labeled development data is used to learn the relationship between shipment characteristics and posted freight rates, then a rate is generated for every row in the final validation dataset. A separate December scenario requires predictions for 31 fixed daily inputs.

Because the final validation period follows the labeled development period, model selection uses a chronological holdout rather than a random split.

2. Dataset Overview

The development file contains 48,000 labeled observations from January 1 through October 31, 2025. The target is **posted_rate**. The final validation file contains 12,000 observations from November 1 through December 31, 2025 and does not include the target.

Feature group	Examples
Identifiers	load_id
Lane / geography	pickup, delivery, pickup_lat/lon, delivery_lat/lon
Shipment	distance, equipment, weight
Market / quote	market_index, quote_signal
Time	date
Target	posted_rate (development only)

3. Data Quality and Cleaning

- Missing numerical values were handled by median imputation inside the modeling pipeline, so imputation statistics are learned from training data only.
- Negative shipment weights were treated as invalid and converted to missing values before imputation; 592 originally missing or invalid weight values were handled this way in the full development data.
- Categorical variables were imputed using the most frequent training value and one-hot encoded.
- OneHotEncoder(handle_unknown='ignore') prevents unseen pickup or delivery locations from causing prediction failures.
- Duplicate rows and load IDs were checked; no duplicate load IDs were present.
- Extreme target rates were retained because they may represent legitimate freight-rate behavior.

4. Exploratory Analysis and Feature Engineering

Distance had the strongest individual numerical relationship with posted_rate, with a Pearson correlation of approximately 0.909.

A Haversine geographic distance was investigated but was almost perfectly redundant with the supplied distance feature (correlation approximately 0.9995), so it was not added to the final model.

The date was transformed into month, day of week, day of month, and day of year. Potential leakage features such as rate_per_mile were excluded because they use the target.

5. Validation Strategy

January–September 2025 was used for training and October 2025 for validation. After selecting the model, it was retrained on all January–October labeled data before predicting November–December.

Partition	Period	Rows
Training	Jan–Sep 2025	43,147
Validation	Oct 2025	4,853
Final model training	Jan–Oct 2025	48,000
Final prediction	Nov–Dec 2025	12,000

6. Model Experiments

The following configurations were compared under the same preprocessing and chronological validation setup.

Model	MAE	RMSE	R^2
Ridge Regression	$170.67	$655.66	0.8160
Random Forest	$219.48	$675.48	0.8047
HistGradientBoosting	$158.40	$661.64	0.8126
Log-target HistGradientBoosting	$116.85	$648.56	0.8200
Tuned Log-target HGB (L2=5)	**$115.45**	**$647.40**	**0.8206**

The log-target approach improved the overall chronological validation results because the freight-rate target is strongly right-skewed. Hyperparameter testing then increased L2 regularization from 1.0 to 5.0. The resulting configuration achieved MAE $115.45, RMSE $647.40, and R^2 0.8206 on the October holdout.

7. Final Model

The selected pipeline was retrained on all 48,000 labeled observations. It uses HistGradientBoostingRegressor with 300 iterations, learning rate 0.05, 31 maximum leaf nodes, L2 regularization of 5.0, and random_state=42. The target is transformed with log1p during training and expm1 for final predictions.

- Numerical median imputation.
- Categorical most-frequent imputation.
- One-hot encoding.
- Calendar feature engineering.
- HistGradientBoosting regression.
- Log-transformed target modeling.

8. Final Validation Predictions

Exactly 12,000 predictions were generated for load IDs TE-000001 through TE-012000. Structural checks confirmed 12,000 rows, unique IDs, no missing predictions, no duplicates, and no non-positive predictions.

Statistic	Predicted rate
Count	12,000
Minimum	$212.36
Median	$2,038.67
Mean	$2,356.14
75th percentile	$3,370.67
95th percentile	$4,897.79
Maximum	$6,574.85
> $5,000	534
> $10,000	0

9. December Fixed-Lane Scenario

The December chart input contains 31 daily observations for Lexington → Fort Wayne, 360 miles, Dry Van, and 32,000 lb. The file does not provide all features used by the model, so pickup and delivery coordinates were recovered from corresponding training records. The unavailable market_index and quote_signal fields were left missing and handled by the pipeline's training-fitted median imputation.

The final daily predictions range from $830.76 to $843.94, with a mean of $839.81.

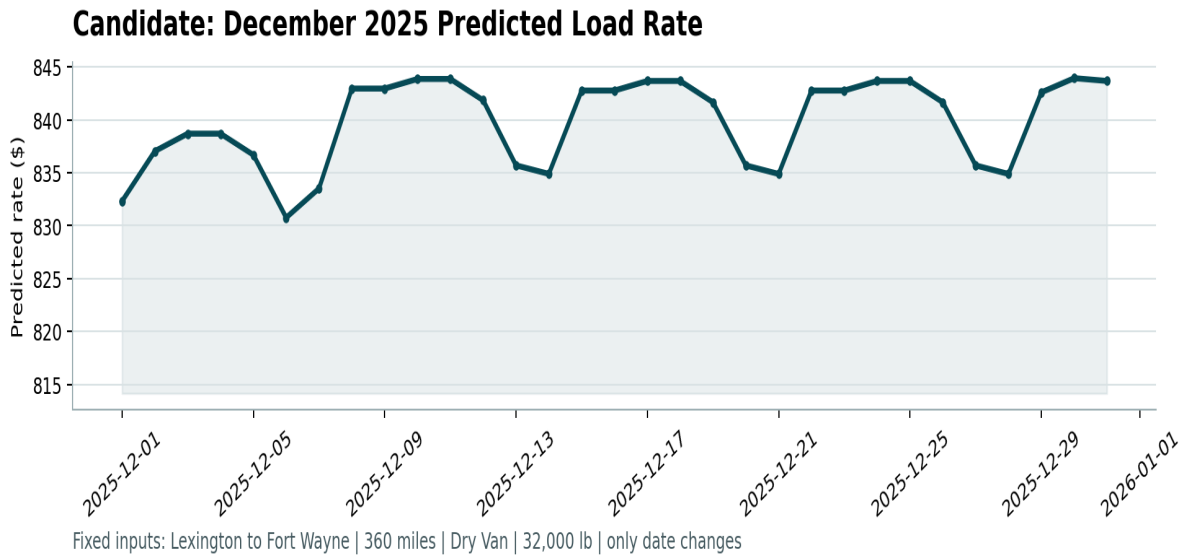


Figure 1. December 2025 daily predictions used for the assessment chart.

10. Error Analysis and Limitations

The October chronological holdout shows strong overall predictive performance, but a small number of extreme high-rate observations remain difficult to predict. Some actual rates above $17,000 were predicted several thousand dollars lower. This indicates that rare extreme-rate behavior is not fully explained by the available shipment, lane, market, quote, and date features.

Because November–December target values are withheld, the true final generalization error cannot be calculated locally. The final evaluation is performed by Spotter after submission.

11. Reproducibility

The project contains the training/prediction script, input datasets, generated prediction files, official score.py checker, December chart, requirements, and this report.

Commands

- pip install -r requirements.txt
- python train_predict.py
- python score.py --predictions validation_predictions.csv --december-predictions december_chart_inputs_filled.csv

12. Conclusion

The final solution combines leakage-aware preprocessing, chronological validation, calendar feature engineering, robust categorical handling, and log-target gradient boosting. The selected final configuration uses HistGradientBoosting with 300 iterations, learning rate 0.05, 31 maximum leaf nodes, and L2 regularization of 5.0.

On the October development holdout, this configuration achieved MAE $115.45, RMSE $647.40, and R^2 0.8206. The official local checker accepted all 12,000 final validation predictions and all 31 December predictions.

The final November–December target values are withheld, so the official generalization performance will be determined by Spotter after submission.